I. Murray-von Neumann equivalence

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Suppose $P_1, P_2 \in B(H)$ are orthogonal projections on two closed subspaces in H.

We may define:

$P_1 \wedge P_2 :=$ projection onto $P_1(H) \cap P_2(H)$

$P_1 \vee P_2 :=$ projection onto $\overline{P_1(H) + P_2(H)}$

$= I - (I - P_1) \wedge (I - P_2)$

↑ If $P_M : H \to M$ for a closed subspace M, then $I - P_M = P_{M^\perp}$, so $(I - P_1) \wedge (I - P_2)$ projects onto $P_1(H)^\perp \cap P_2(H)^\perp$, the orthogonal complement of which is $\overline{P_1(H) + P_2(H)}$.

Recall: $P_\alpha \geq P_\beta \iff P_\beta(H) \subset P_\alpha(H)$

$(\Rightarrow)$ $P_\alpha \geq P_\beta \Rightarrow \ker(P_\alpha) \subset \ker(P_\beta)$

(If $P_\alpha x = 0$, then $\langle P_\beta x, x \rangle \leq 0$ implying $P_\beta x = 0$.)

$\Rightarrow P_\beta(H) = \ker(P_\beta)^\perp \subset \ker(P_\alpha)^\perp = P_\alpha(H)$.

$(\Leftarrow)$ If $\ker(P_\alpha) \subset \ker(P_\beta)$, then

$\forall x \in \operatorname{im} P_\alpha : \langle (P_\alpha - P_\beta) x, x \rangle = 0$

$$\forall x \in \underbrace{(\ker P_\alpha)^\perp}_{=\operatorname{im}(P_\alpha)} : \langle (P_\alpha - P_\beta)x, x \rangle = \langle x - P_\beta x, x \rangle \geq 0.$$

This implies $P_1 \wedge P_2 \leq P_1, P_2 \leq P_1 \vee P_2$.
(Also, shows that $\leq$ is a partial order on $\mathcal{P}(H)$.)

More generally, if $(P_\alpha)_{\alpha \in I}$ is a family of projections in $B(H)$, we may define

$$\bigwedge_\alpha P_\alpha = \text{projection onto } \bigcap_\alpha P_\alpha(H)$$

$$\bigvee_\alpha P_\alpha = I - \bigwedge_\alpha (I - P_\alpha)$$

$$\left(= \text{projection onto } \overline{\sum_\alpha P_\alpha(H)}\right)$$

<u>Prop.</u> If $A \subset B(H)$ is a von Neumann algebra then the set of projections

$$\mathcal{P}(A) := \{T \in A ; T^* = T^2 = T\}$$

is a complete lattice (i.e. any subset admits an infimum and a supremum) w.r.t. the partial order $\leq$.

<u>Pf.</u> We want to show $\bigwedge_\alpha P_\alpha, \bigvee_\alpha P_\alpha \in A$ $(= A'')$ (double commutant thm.) for any $(P_\alpha)_{\alpha \in I} \subset \mathcal{P}(A)$. Consider first $P_1, P_2 \in \mathcal{P}(A)$. Given $T \in A'$, we have

$$P_i T = T P_i \quad (\text{and } P_i T^* = T^* P_i) \qquad (i=1,2)$$

which is equivalent to saying that $P_i(H)$ and $P_i(H)^\perp$ are inv. subspaces for T, T^*. Now

$$T(P_1(H) \cap P_2(H)) \subset T(P_1(H)) \cap T(P_2(H)) \subset P_1(H) \cap P_2(H)$$

i.e. $P_1(H) \cap P_2(H)$ is invariant for T (and similarly, for T^*, implying $(P_1(H) \cap P_2(H))^\perp$ is invariant for T), so

$$(P_1 \wedge P_2) T = T (P_1 \wedge P_2),$$

implying $P_1 \wedge P_2 \in A'' = A$. Since $I \in A$, we also have $P_1 \vee P_2 = I - (I - P_1) \wedge (I - P_2) \in A$.

That $P_1 \wedge P_2$ and $P_1 \vee P_2$ are indeed the greatest lower and least upper bound of $\{P_1, P_2\}$ is clear from their images.

The same argument works for $(P_\alpha)_{\alpha \in I}$. $\boxtimes$

Def. $P_1, P_2 \in \mathcal{P}(A)$ are <u>(Murray–von Neumann) equivalent</u> if there exists an operator $V \in A$ s.t.

$$P_1 = V^* V \quad \text{and} \quad P_2 = V V^*.$$

(Denote $P_1 \sim_A P_2$ and call them "equivalent through V".)

Remark. In this case, there holds $\forall x \in H$:

$$\|Vx\|^2 = \langle \overbrace{V^*V}^{P_1} x, x \rangle = \begin{cases} \|x\|^2 & ; x \in P_1(H) \\ 0 & ; x \in P_1(H)^\perp \end{cases}$$

and $$\|V^*x\| = \begin{cases} \|x\| & ; x \in P_2(H) \\ 0 & ; x \in P_2(H)^\perp . \end{cases}$$

We call operators V and V^* <u>partial isometries</u>, i.e. $V \in B(H)$ is such that $\|Vx\| = \|x\|$ $\forall x \in \ker(V)^\perp$.

In our case V is a partial isometry with

$\ker(V)^\perp = P_1(H)$ and $\operatorname{im}(V) = \ker(V^*)^\perp = P_2(H)$.

"initial space" "final space"

Intuitively, V maps $P_1(H)$ isometrically onto $P_2(H)$.

It's easy to see that $\sim_A$ is indeed an equiv. rel.

To establish some basic properties of $\sim_A$, we need:

Theorem. (Polar decomposition in von Neumann algebras)
Let $A \subset B(H)$ be a von Neumann algebra.
If $T \in A$, then the operators $V, S \in B(H)$ of the polar decomposition

$$T = VS \quad \left(\begin{array}{l} V \ldots \text{partial isometry} \\ S \ldots \text{positive operator} \end{array}, \ \ker V = \ker(S) \right)$$

belong to A.

Remark. Recall that we construct the decomposition in $B(H)$ as

$V_0 : \operatorname{im}(S) \to H$
$Sx \mapsto Tx$

$$S := \sqrt{T^*T}, \quad V(x) := \begin{cases} T(V_0(x)) & ; x \in \overline{\operatorname{im}(S)} \\ 0 & ; x \in \overline{\operatorname{im}(S)}^\perp \end{cases}$$

and that it is unique.

<u>Pf.</u> Since $S = \sqrt{T^*T}$, we immediately have $S \in A$.

$V \in A$. Let $U \in A'$. $\forall x \in H$,

$$UVSx = UTx = TUx = VSUx = VUSx,$$

implying $UV = VU$ holds on $\overline{\operatorname{im}(S)}$. But

$$\overline{\operatorname{im}(S)} \underset{S \geq 0}{=} (\ker S)^\perp = (\ker V)^\perp$$

So it suffices to show $UV = VU$ on $\ker(V)$.

If $x \in \ker(V)$, then

$UVx = 0$ and

$VUx = 0$

↳ WTS: $U(\ker V) \subset \ker V$ or equiv.

$U(\ker S) \subset \ker S$

But $SUx = USx = 0$ $\forall x \in \ker S$.

Hence $V \in A'' = A$. ∎

<u>Corollary.</u> If $T \in A$ and P_T, P_{T^*} denote projections onto $\overline{\operatorname{im}(T)}$, $\overline{\operatorname{im}(T^*)}$, then $P_T \sim_A P_{T^*}$

<u>Pf.</u> First recall that since $T, T^* \in A$, $P_T, P_{T^*} \in A$ (because of $T^{\frac{1}{n}} \xrightarrow[SOT]{n} P_T$ trick).

Now $T = VS$ as above and V is a partial isometry,

$$\ker(V)^\perp = \ker(S)^\perp \underset{S = \sqrt{T^*T}}{=} \ker(T)^\perp = \overline{\operatorname{im} T^*} = P_{T^*}(H)$$

$$\overline{\mathrm{im}(V)} = \overline{\mathrm{im}(T)} = P_T(H),$$

and so $P_{T^*} = V^*V$, $P_T = VV^*$, i.e.

$P_{T^*} \sim_A P_T$. $\blacksquare$

A corollary of this corollary rests upon the following observation:

If $P_1 \geq P_2$, then $P_1 - P_2$ is also a projection, onto $P_1(H) \cap P_2(H)^\perp$.

$\hookrightarrow$ $P_1 \geq P_2 \iff P_2(H) \subset P_1(H)$,

hence $P_1 P_2 = P_2 = P_2 P_1$, and so

$$(P_1 - P_2)^2 = P_1^2 - P_1P_2 - P_2P_1 + P_2^2$$
$$= P_1 - P_2 - P_2 + P_2 = P_1 - P_2.$$

The image of $P_1 - P_2$ is easy to establish.

<u>Thm.</u> (Kaplansky's formula)

For $P_1, P_2 \in \mathcal{P}(A)$, there holds

$$(P_1 \vee P_2) - P_2 \sim_A P_1 - (P_1 \wedge P_2).$$

So: $\overline{M_1 + M_2} \cap M_2^\perp \xrightarrow[\text{isometry}]{\exists} M_1 \cap \overline{M_1^\perp + M_2^\perp}$

<u>Prf.</u> Let's show:

$$(P_1 \vee P_2) - P_2 = P_{(I-P_2)P_1}$$

$$\hookrightarrow \quad \overline{\operatorname{im}(I-P_2)P_1} = \left(\ker\left((I-P_2)P_1\right)^*\right)^\perp$$

$$= \left(\ker P_1^*(I-P_2)^*\right)^\perp = \left(\ker P_1(I-P_2)\right)^\perp$$

$$= \left(\ker(I-P_2) + \operatorname{im}(I-P_2) \cap \ker(P_1)\right)^\perp$$

$(A+B)^\perp = A^\perp \cap B^\perp$

$$= \left(P_2(H) + \left(P_2(H)^\perp \cap P_1(H)^\perp\right)\right)^\perp$$

$$= P_2(H)^\perp \cap \overline{\left(P_2(H) + P_1(H)\right)}$$

$$P_1 - (P_1 \wedge P_2) \underset{\text{similarly}}{=} P_{P_1(I-P_2)} = P_{((I-P_2)P_1)^*}$$

By the corollary, we are done. $\boxtimes$

<u>Proposition</u>. If $(P_\alpha)_{\alpha\in I}, (Q_\alpha)_{\alpha\in I} \subset \mathcal{P}(A)$ are two nets of projections with

- $P_\alpha(H) \perp P_\beta(H)$
- $Q_\alpha(H) \perp Q_\beta(H)$

$\Big\}\ \forall \alpha \neq \beta$ each of the two nets is orthogonal

- $P_\alpha \sim_A Q_\alpha \quad \forall \alpha \in I$,

then

$$P = \sum_{\alpha\in I}^{SOT} P_\alpha \sim_A Q = \sum_{\alpha\in I}^{SOT} Q_\alpha .$$

Recall: $\sum_{\alpha\in I}^{SOT} P_\alpha = \lim_{\substack{F\subset I\\ \text{finite}}}^{SOT} \Big(\underbrace{\sum_{\alpha\in F} P_\alpha}\Big)$

if $F\subset F'$, then $\sum_{\alpha\in F} P_\alpha \leq \sum_{\alpha\in F'} P_\alpha \Rightarrow \big(\sum_{\alpha\in F} P_\alpha\big)_F$ increasing

and $\|Px\|^2 = \lim_{\substack{F\subset I\\ \text{finite}}} \sum_{\alpha\in F} \|P_\alpha x\|^2$.

Here, P is the projection onto the space $\overline{\sum_{\alpha\in I} P_\alpha(H)}$, hence $P = \bigvee_{\alpha\in I} P_\alpha$.

So the proposition says: $\bigvee_\alpha P_\alpha \sim_A \bigvee_\alpha Q_\alpha$.

<u>Pf.</u> $P_\alpha = V_\alpha^* V_\alpha$, $Q_\alpha = V_\alpha V_\alpha^*$

where P_α and Q_α are equivalent through $V_\alpha \in A$. Define:

$$V_F := \sum_{\alpha\in F} V_\alpha \rightsquigarrow V := \lim_{\substack{F\subset I\\ \text{finite}}}^{SOT} V_F$$

↳ Convergence?

Define $M = \sum_{\alpha\in I} P_\alpha(H)$ and $\tilde{V}: M\to M$,

$\tilde{V}x = \sum_{\alpha\in F} V_\alpha x_\alpha$, where $x = \sum_{\alpha\in F} \underbrace{x_\alpha}_{\in P_\alpha(H)}$

Extending $\tilde{V}$ to $\bar{M}$, we find $\tilde{V} = V$.

Since $A = \bar{A}^{(SOT)}$, $V\in A$.

We get $V^*V = \lim\limits_{\substack{F\subset I\\ \text{finite}}} \sum\limits_{\alpha\in F} V_\alpha^* \cdot \lim\limits_{\substack{F'\subset I\\ \text{finite}}} \sum\limits_{\alpha'\in F'} V_{\alpha'}$

$$= \lim_{\substack{F,F'\subset I\\ \text{finite}}} \underbrace{\sum_{\alpha} V_\alpha^* \cdot \sum_{\alpha'} V_{\alpha'}}_{V_\alpha^* V_{\alpha'} = 0 \text{ if } \alpha\neq\alpha' \text{ by } \perp} = \lim_{\substack{F\subset I\\ \text{finite}}} \sum_{\alpha\in F} \underbrace{V_\alpha^* V_\alpha}_{P_\alpha}$$

$= P$. Similarly, $VV^* = Q$. $\blacksquare$